Graph-Based Analysis of Cascading Load Shedding in Power Grid Networks

Your Name
Graph Mining Final Project

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Abstract

This project studies cascading failure behavior in power grid networks using graph mining and power-flow simulation. Using the IEEE 118-bus and IEEE 300-bus test systems from pandapower, we constructed graph representations of each grid, computed node and edge centrality metrics, and compared graph-based importance to simulation-based cascade criticality. Cascades were modeled through iterative AC power-flow simulations where overloaded lines were removed and impacts were measured using cascade size, load shed percentage, and largest connected component size. Results show that graph metrics alone do not fully explain cascade severity, especially in case118, but node-level metrics such as degree and node betweenness become stronger predictors in the larger case300 network. Overall, the project shows that graph mining can help screen for vulnerable components, but simulation-based validation is still necessary for understanding true cascade impact.

1 Introduction

Power grid reliability is an important infrastructure problem because the failure of one component can trigger a chain reaction of additional failures. From a graph mining perspective, a power grid is naturally represented as a network, where buses correspond to nodes and transmission lines correspond to edges. This makes it possible to apply centrality measures such as degree, betweenness, closeness, and eigenvector centrality to identify structurally important components.

However, structural importance does not always imply physical criticality. A line or bus may appear important in the graph but may not cause large load shedding when removed. Similarly, a less central component may still trigger a severe cascade because of the physical flow conditions of the grid. This project investigates the relationship between graph-based importance and simulation-based cascade criticality. The main research questions are:

1. Which nodes and edges are most critical under cascade dynamics?
2. How well do graph metrics predict simulation-based criticality?
3. How do overload thresholds affect cascade severity?
4. Do the results generalize from the IEEE 118-bus network to the larger IEEE 300-bus network?

2 Background

2.1 Power Grids as Graphs

A power grid can be modeled as a graph $G = (V, E)$, where each bus is represented as a node in V and each transmission line is represented as an edge in E. This graph representation captures the physical connectivity of the grid and allows graph mining techniques to be applied to power system vulnerability analysis.

In this project, two benchmark networks were used: the IEEE 118-bus system and the IEEE 300-bus system. These networks were loaded using the pandapower library. The graph representation was created from the bus and line tables, while the original pandapower network was used for AC power-flow simulation.

2.2 Graph Centrality Metrics

Several graph metrics were computed to measure different forms of structural importance. **Degree** measures the number of direct connections incident to a node. A high-degree bus is locally well-connected. **Node betweenness centrality** measures how often a node lies on shortest paths between other nodes. A high-betweenness bus may act as a bridge between different regions of the grid. **Edge betweenness centrality** measures how often an edge lies on shortest paths between node pairs. A high-betweenness line may represent a structurally important connection. **Closeness centrality** measures how close a node is to all other nodes in the network. **Eigenvector centrality** assigns higher scores to nodes connected to other important nodes. These metrics capture topological importance, but they do not directly account for electrical flow, generation and load placement, line capacity, or AC power-flow constraints.

2.3 Cascading Failures and Load Shedding

A cascading failure begins with an initial outage, such as the removal of one transmission line or the removal of all lines incident to a bus. After the initial contingency, power flow is recomputed. If additional lines become overloaded, those lines are tripped and the process repeats until no new overloads occur or the solver fails. The impact of a cascade was measured using several quantities:

- **Cascade size:** the number of lines tripped during the cascade.
- **Load shed percentage:** the percentage of total demand that is no longer served.
- **Largest connected component size:** the size of the largest remaining connected component after the cascade.
- **Solver failure status:** whether the AC power-flow solver failed during the simulation.

3 Methodology

3.1 Data

The experiments were performed on the IEEE 118-bus and IEEE 300-bus networks. Both networks were accessed through pandapower.

Table 1: Network summary for case118 and case300.

Dataset	Buses	Lines	Graph Edges	Description
case118	118	173	166	IEEE 118-bus system
case300	300	283	281	IEEE 300-bus system

The difference between the number of physical lines and graph edges occurs because the graph representation collapses some duplicate or parallel connections into a single graph edge.

3.2 Graph Construction

The IEEE 118-bus and 300-bus benchmark systems were loaded using pandapower, which builds on standard power-system modeling tools and MATPOWER-style test cases [3, 4]. Graph metrics were computed using NetworkX [5], and visualizations were produced with Matplotlib [6].

3.3 Cascade Simulation Pipeline

The cascade simulation followed an iterative process:

1. Run a baseline AC power-flow calculation.
2. Remove an initial line or remove all lines incident to an initial bus.
3. Re-run AC power flow.
4. Identify overloaded lines.
5. Trip overloaded lines.
6. Repeat until no new overloads occur or the solver fails.
7. Record cascade size, load shed, largest connected component size, and solver status.

A line was considered overloaded if its loading exceeded a baseline-relative threshold:

$$\text{threshold}_i = \max(\text{baseline loading}_i, \text{minimum baseline loading}) \times (1 + \alpha)$$

where α controls the overload margin. This baseline-relative threshold was used because the original networks were not stressed enough for a simple absolute overload rule to generate meaningful cascades. Unless otherwise stated, the main reported simulations used $\alpha = 0.2$ and minimum baseline loading $= 0.1$.

3.4 Edge-Level Experiments

For the edge-level experiments, each transmission line was removed one at a time as the initiating contingency. After each initial line removal, the cascade simulator was run and the resulting impact was recorded. Edge betweenness was then compared against load shed percentage and cascade size.

3.5 Node-Level Experiments

For the node-level experiments, each bus was treated as the initiating contingency by removing all lines incident to that bus. The resulting cascade impact was then compared against node-level graph metrics such as degree, node betweenness, closeness, and eigenvector centrality.

3.6 Threshold Sensitivity Analysis

To test how sensitive the cascade results were to model calibration, the overload margin α and the minimum baseline loading parameter were varied. For each parameter setting, the mean load shed, maximum load shed, mean cascade size, maximum cascade size, and metric correlations were recorded.

4 Results and Discussion

4.1 Centrality Analysis

The first part of the analysis identified the most important buses and lines according to graph centrality. In case118, several buses appeared in both the top degree and top betweenness rankings, including buses 48, 76, 79, and 99. This suggests that these buses were both highly connected and important as bridges in the network.

For case300, the largest node betweenness values were much smaller than in case118. This suggests that shortest-path dependence was more distributed across the larger network.

Table 2: Top critical graph components by betweenness.

Dataset	Component	Betweenness	Additional Information
case118	Bus 76	0.358	Degree 6
case118	Line 109, Bus 68–76	0.273	Highest edge betweenness
case300	Bus 40	0.035	Degree 3
case300	Line 73, Bus 40–60	0.033	Highest edge betweenness

4.2 Worst Cascade Results

The worst edge-triggered cascades were identified by sorting the simulations by load shed percentage. In case300, the most severe line-triggered contingency was line 225, which produced approximately 7,564 MW of shed load and 31.72% load shed.

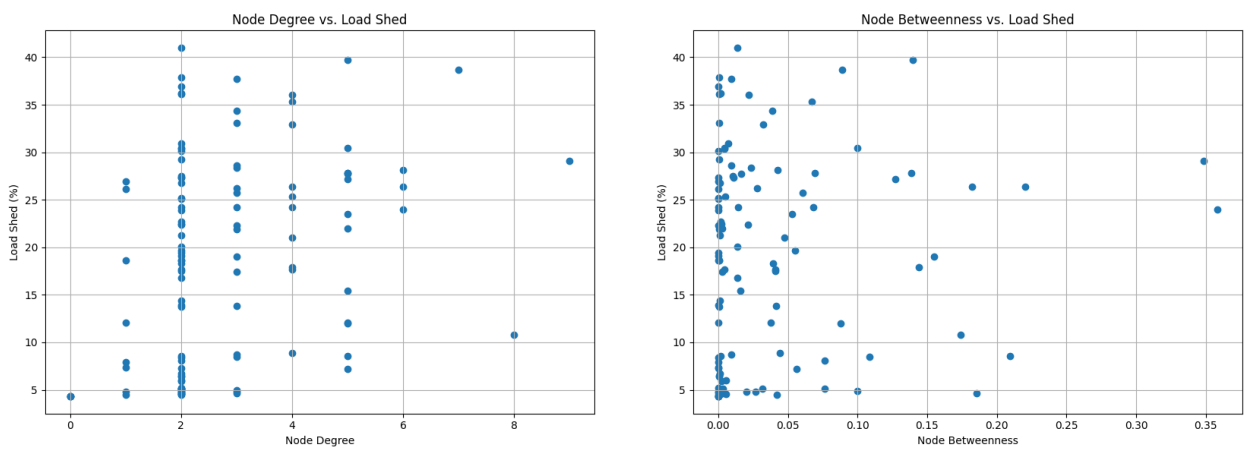
Table 3: Top 10 worst edge-triggered cascades for case300.

Initial Line	Cascade Size	LCC Size	Shed Load MW	Shed Percent
225	79	30	7564	31.72
201	64	28	7000	29.35
147	66	28	6859	28.77
228	69	34	6264	26.27
269	74	34	6261	26.26
263	42	68	5840	24.49
177	74	25	5801	24.33
99	81	34	5728	24.02
262	46	40	5703	23.92
214	36	80	5688	23.85

The largest cascades were not always caused by the components with the highest betweenness. This suggests that physical operating conditions and local network structure can be more important than shortest-path topology alone.

4.3 Node Metrics vs. Simulation Impact

The node-level results compared graph metrics against simulation impact. The scatter plots show that degree had a stronger relationship with load shed than node betweenness in case118, although neither metric fully explained cascade severity.



(a) Node degree vs. load shed.

(b) Node betweenness vs. load shed.

Figure 1: Case118 node metrics compared against simulation impact.

For case118, node degree was a stronger predictor of simulation impact than node betweenness. However, the plots also show important exceptions: some low-degree buses still caused large load shed, while some high-betweenness buses produced only moderate impact. This supports the conclusion that graph metrics are useful but incomplete.

4.4 Correlation Summary

The correlation analysis gives a quantitative comparison between graph metrics and simulation impact. Pearson correlation measures the strength of a linear relationship, while Spearman correlation measures whether the ranking of graph importance agrees with the ranking of simulation impact.

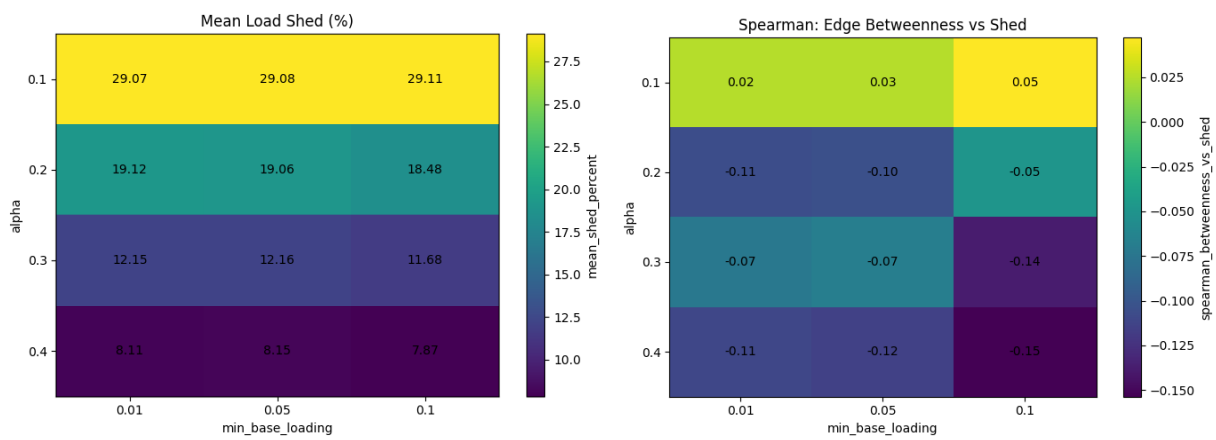
Table 4: Correlation between node-level graph metrics and simulation impact. (D = Degree and NB = Node Betweenness)

Dataset	Metric	Pearson vs. Shed	Spearman vs. Shed	Pearson vs. Cascade	Spearman vs. Cascade
118	D	0.273	0.306	0.406	0.556
118	NB	0.072	0.179	0.133	0.287
300	D	0.406	0.747	0.545	0.794
300	NB	0.124	0.626	0.282	0.648

The results show a clear difference between the two networks. In case118, graph metrics had weak to moderate correlation with cascade impact. Degree was the best predictor, but its correlation was still not strong enough to fully explain cascade severity. In case300, the correlations were substantially stronger across all node metrics. Degree had the highest correlation with both load shed and cascade size, suggesting that graph structure was more predictive in the larger network.

4.5 Threshold Calibration Results

The threshold calibration analysis tested whether the cascade behavior depended strongly on the overload threshold parameters. The main parameter was α, where a larger α means a larger overload margin and therefore a less sensitive cascade model.



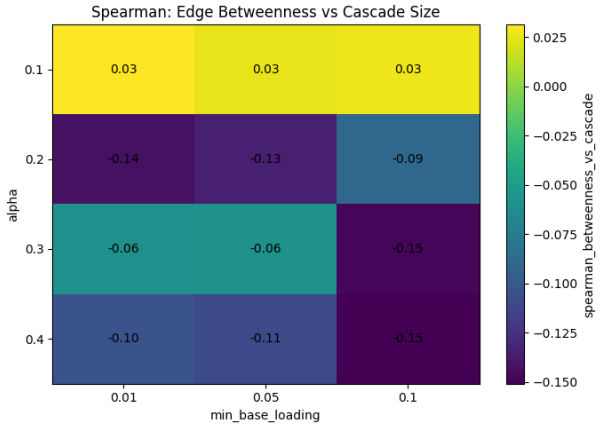


Figure 2: Case118 threshold calibration heatmaps.

In case118, increasing α reduced mean load shed. This makes sense because a larger overload margin means fewer lines are classified as overloaded. However, the correlation between edge betweenness and cascade impact remained weak across most threshold settings. This suggests that edge betweenness was not a strong standalone predictor of edge-triggered cascade severity.

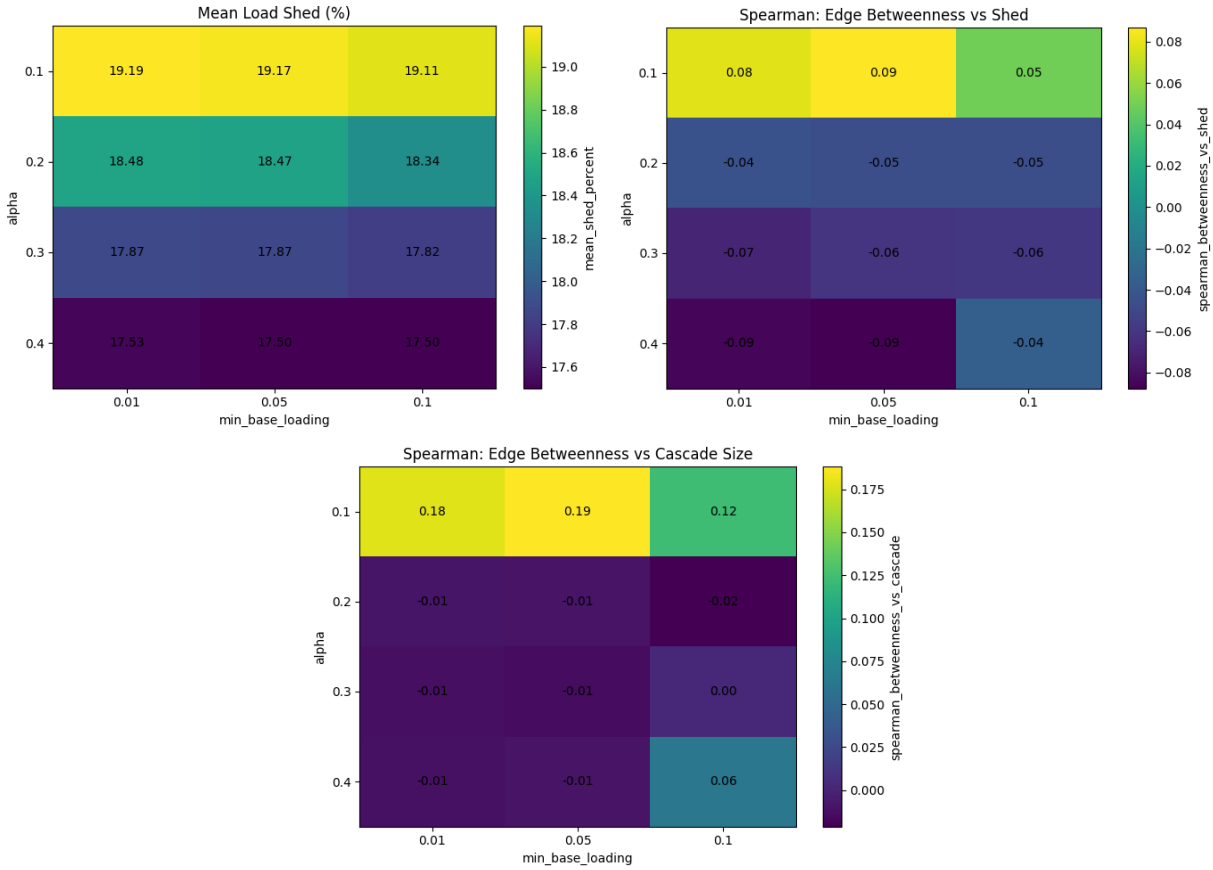


Figure 3: Case300 threshold calibration heatmaps.

Case300 showed a similar trend: increasing α reduced the average load shed. However, the edge betweenness correlations were still weak or near zero in many settings. One reason edge betweenness performed weakly is that it is a purely topological shortest-path metric. Edge betweenness counts how often a transmission line lies on shortest paths between pairs of buses, but power-grid flows do not move only along shortest paths. In an electrical network, power distributes across many available paths according to impedance, Kirchhoff's laws, and the operating state of the system. Therefore, a line can have high edge betweenness in the graph-theoretic sense without carrying a large amount of redistributed power after a contingency. Conversely, a line with low edge betweenness may still be important if its removal forces power onto already heavily loaded nearby lines. This helps explain why edge betweenness was not a reliable predictor of load shed or cascade size in our simulations. This reinforces the idea that edge-level shortest-path importance does not reliably predict which line outages produce the worst cascades.

This agrees with the argument made by Bompard et al., who emphasize that purely topological metrics can be misleading for power-grid vulnerability because they ignore the physical laws governing electrical flow. [2] Our results show the same pattern: edge betweenness sometimes identifies structurally central lines, but simulation-based criticality depends on how power is redistributed after the line is removed.

4.6 Comparison Between case118 and case300

Table 5: Summary comparison between case118 and case300.

Quantity	case118	case300
Buses / Lines	118 / 173	300 / 283
Graph Edges	166	281
Max Load Shed	45.2%	31.7%
Max Cascade Size	109 lines	85 lines
Best Node Predictor	Degree	Degree
Top Node Betweenness Bus	Bus 76	Bus 40
Top Edge Betweenness Line	Line 109	Line 73

The main difference between the two networks is that node-level graph metrics were much more predictive in case300. In case118, graph-based importance and simulation-based criticality only partially matched. In case300, the stronger correlations suggest that topology played a larger role in determining cascade impact.

At the same time, edge betweenness was weak in both networks for predicting edge-triggered cascade severity. This was one of the most important findings of the project because it shows that the most central edge in a graph is not necessarily the edge whose failure causes the most load shedding.

5 Challenges and Adjustments

One major challenge was choosing an overload rule that produced meaningful but not unrealistic cascades. The default networks were not heavily stressed enough for a simple absolute overload threshold to produce interesting results. In early experiments, many contingencies produced weak or no cascades.

To address this, we switched from an absolute threshold to a baseline-relative threshold. This made the overload rule depend on how much each line changed relative to its original operating condition. I also added a minimum baseline loading parameter to avoid instability from lines with near-zero baseline loading.

Another challenge was solver failure. In some severe contingencies, the AC power-flow solver failed after parts of the network were disconnected or after too many lines were tripped. These cases were recorded as part of the simulation output rather than ignored.

A final challenge was interpreting graph metrics in a physical system. Graph metrics are based mostly on topology, while power grids are governed by electrical constraints. This means that centrality can help identify important components, but it cannot fully replace simulation.

6 Conclusions

The difference between case118 and case300 suggests that the usefulness of graph metrics depends on the structure and operating characteristics of the network. In case118, the relationship between graph metrics and cascade impact was weaker, meaning that local operating conditions and post-contingency redistribution effects were more important than simple topology. Several low-degree buses still produced large load shed, showing that a component does not need to be highly connected to be operationally critical.

In case300, the correlations between node-level graph metrics and simulation impact were much stronger. This suggests that, in the larger network, the graph structure aligned more closely with the way cascading failures propagated. Degree, closeness, eigenvector centrality, and node betweenness all became more predictive of load shed and cascade size. It is important to note that even a correlation of 0.747 (degree vs. shed, our highest correlation) still leaves a lot of variance unexplained. Edge betweenness remained weak, which reinforces the idea that shortest-path-based edge importance does not fully represent electrical power flow.

Overall, the project shows that graph mining can help identify potentially critical grid components, but purely topological metrics are not enough on their own. The strongest future direction is to combine graph-based features with electrical operating information such as line loading, impedance, capacity margins, generator placement, and load placement.

6.1 Connection to Related Work

Our project is closely related to prior work studying whether graph-theoretic metrics can identify vulnerable components in power grids. Hines, Cotilla-Sanchez, and Blumsack study this question directly by comparing purely topological vulnerability measures with a power-flow-based cascading failure model. Their main conclusion is that topological metrics can reveal general trends, but they can also be misleading when used in isolation because power grids are governed by electrical constraints such as Ohm's law and Kirchhoff's laws rather than only shortest-path connectivity.

This connects strongly to our results. In our experiments, metrics such as degree, node betweenness, closeness, and eigenvector centrality were compared against simulation-based outcomes such as load shed percentage and cascade size. The results showed that graph metrics were not equally reliable predictors of failure impact. For example, in case118, node degree was more informative than node betweenness, but neither metric fully explained the most damaging cascades. This agrees with the broader conclusion from Hines et al.: purely topological centrality can point in the right direction on average, but it does not fully capture real grid vulnerability.

A second related direction is the work by Bompard, Pons, and Wu on extended topological metrics for power grid vulnerability. Instead of relying only on standard graph metrics, they

propose incorporating electrical properties such as power transfer distribution, line flow limits, bus type, and electrical distance into centrality measures. [2] Their extended betweenness metric is designed to bridge the gap between pure graph theory and full power-flow analysis. This is especially relevant to our project because our results show a clear gap between graph-based importance and simulation-based criticality.

Together, these papers suggest a clear extension of our work. Rather than relying only on purely topological graph metrics, future versions of this project could use electrically-informed graph metrics. For example, weighted betweenness could include line reactance or impedance, edge weights could reflect thermal limits or capacity margins, and power-transfer distribution factors could be used to estimate which lines are electrically important after a contingency.

This would move the graph mining portion of the project closer to the actual physics of the grid. In this sense, electrically-informed metrics would not replace full power-flow simulation, but could serve as a faster screening tool for identifying components that deserve more detailed simulation.

7 Future Work

The main future direction is not simply to run more simulations, but to make the graph representation more physically meaningful. Our next step would be to combine graph-based features with electrical operating features such as line loading, capacity margins, impedance, generator placement, and load placement. This would allow future predictive models to use both topology and physical operating conditions when estimating cascade severity.

Additional future work could validate these hybrid metrics against more detailed simulations. For example, the project could compare AC and DC power-flow cascade models, test additional benchmark networks, include generator redispatch, or evaluate mitigation strategies such as reinforcing high-risk lines. These extensions would help determine whether electrically-informed graph metrics can improve over purely topological metrics while still remaining more efficient than exhaustive contingency simulation.

A final direction would be to compare our results more quantitatively against prior research on power-grid cascading failures and centrality-based vulnerability analysis, especially studies that test whether centrality metrics can reliably predict simulated cascade impact.

References

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